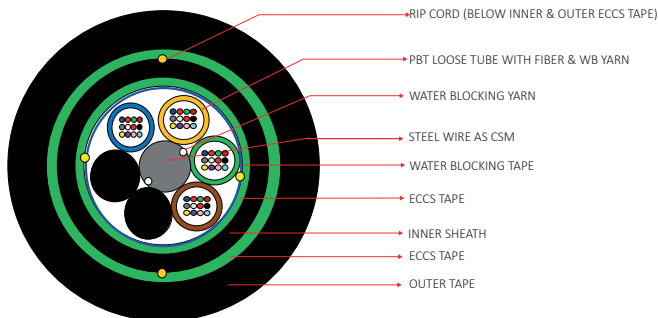
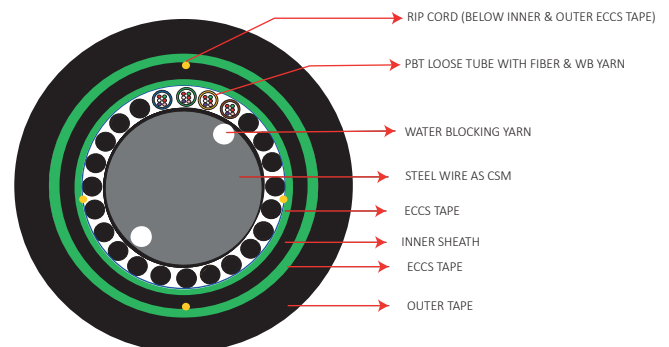


06F-144F DOUBLE METALLIC ARMOURED DOUBLE SHEATH OUTDOOR



12F- TUBE CONSTRUCTION



06F- TUBE CONSTRUCTION

DESCRIPTION

The structure of DATACOM optical cable is to insert single-mode or multi-mode optical fiber into a loose tube made of high-modulus plastic, and the tube is filled with water-blocking compound. Loose tube cables are the product of choice as the backbone in Outside Plant (OSP) environments. The durable loose tube design offers reliable transmission performance over a broad temperature range. Optical fibers are placed inside filled buffer tubes containing the gel. The core is constructed by stranding the buffer tubes around a central member using a reverse oscillating lay (ROL). The core is wrapped with flexible strength members and covered with a water-blocking tape. A corrugated aluminium tape is applied and encased with a black inner jacket. More water-blocking yarns, a corrugated steel armor and a black PE outer jacket complete the cable construction.

APPLICATION

- This cable can be used for outdoor data communications connections including CATV, WAP, CCTV, telecom trunk and access lines.
- These cables are armoured for direct burial and conditions with risk of severe rodent attacks.
- This type of cable can be used as distribution and feeder cables
- Local loop, metro, long-haul and broadband network.

STANDARDS COMPLIANCE

- TIA 568D.3
- ITUG652.D
- ROHS

FEATURES

- Precisely controlling the excess length of the optical fiber ensures that the optical cable has good tensile properties and temperature characteristics.
- The loose tube material itself has good hydrolysis resistance and high strength. The tube is filled with special grease to provide critical protection for the optical fiber.
- Good pressure resistance and softness;
- The IPE sheath has good UV radiation resistance and environmental stress cracking resistance.
- Take the following measures to ensure the water proof performance of optical cables: A single steel wire center reinforcement, special fiber past filled in the loose tube, 100% cable core filling, double-sided plastic-coated steel strip (PSP) improve the moisture-proof ability of the optical cable, and good water-blocking materials prevent longitudinal water seepage of the optical cable.
- Bending radius: static 12.5D, dynamic 25D
- Applicable: pipeline, overhead, direct buried.

06F-144F DOUBLE METALLIC ARMORED DOUBLE SHEATH OUTDOOR CABLE

CONSTRUCTION DETAILS

No of Fibre/Tube	06F per Tube - Blue, Orange, Green, Brown, Slate/Grey & White 12F per Tube - Blue, Orange, Green, Brown, Slate/Grey, White, Red, Black, Yellow, Violet, Pink, Aqua.
Loose Tube	PBT Loose Tube Filled With Water Blocking Yarn
No of Loose Tube	FOR 06F - 1 Loose Tube - Blue & 5 PE Black Fillers FOR 48F - 4 Loose Tubes - Blue, Orange, Green, Brown & 2 PE Black Fillers FOR 96F - 8 Loose Tubes - Blue, Orange, Green, Brown, Slate, White, Red & Black
Central Strength Member	Steel Wire as CSM
Moisture Barrier	Water Swellable Yarn & Water Blocking tape
Core Wrapping	Core Wrapped With Water Blocking Tape & Binder Yarn
Armouring	Corrugated ECCS Tape Armour Over Core
Inner Sheath	HDPE UV Black - 1.2 mm (Nominal)
Armouring	Corrugated ECCS Tape Armour Over Inner PE
Rip cord	Two Rip cord provided below the 1st Armour & Two Rip cords provided below the 2nd Armour
Outer Jacket	HDPE UV & Anti Rodent BLACK Colour - 1.7 mm (Nominal)

FIBER CHARACTERISTICS

Attenuation (Transmission Characteristics)	
@ 1310 nm	≤ 0.36 (dB/Km)
@ 1550 nm	≤ 0.23 (dB/Km)

GEOMETRICAL CHARACTERISTICS

Mode Field Diameter @ 1310 nm	9.2 ± 0.4 μ m
Mode Field Diameter @ 1550 nm	10.4 ± 0.5 μ m
Cladding Diameter	125 ± 0.7 μ m
Cladding Non Circularity	$\leq 1\%$
Core Clad Concentricity Error	≤ 0.5 μ m
Coating Diameter	245 ± 5 μ m
Coating/Cladding Concentricity	≤ 12 μ m

CUT OFF WAVELENGTH

Fibre cut-off Wavelength	≤ 1320 nm
Cable Cut-off Wavelength	≤ 1260 nm

MECHANICAL CHARACTERISTICS

Operating Temperature	-20° C to +70° C
Fibre Proof Test	1%
Stripability Force	$1.3 < F < 8.9$ N
Fibre Curl	≥ 4 meter radius of curvature

DISPERSION

A. Total Dispersion (Chromatic Dispersion)	
1285-1330 nm	≤ 3.5 ps/nm.km
1270-1340 nm	≤ 5.3 ps/nm.km
1550 nm	≤ 18.0 ps/nm.km
1625 nm	≤ 22.0 ps/nm.km
B. Polarization Mode Dispersion at 1310 & 1550 nm	
PMD (Max. Individual)	≤ 0.2 ps/sqrt.km
C. Dispersion Slope & Wave Length	
Zero Dispersion Wavelength	1300-1324 nm
Zero Dispersion Slope	≤ 0.092 ps/nm ² .km

06F-144F DOUBLE METALLIC ARMORED DOUBLE SHEATH OUTDOOR CABLE

CABLE MECHANICAL & PHYSICAL CHARACTERISTICS

CABLE MECHANICAL / PHYSICAL / ENVIRONMET CHARACTERISTICS

Max. Tensile Strength (IEC 60794-1-2 E1)	2700 N (Short Term) 800 N (Long Term)	Cable Diameter (Nominal)	For 06F -15.0 mm $\pm$ 5% For 48F -15.0 mm $\pm$ 5% For 96F -16.5 mm $\pm$ 5%
		Nominal Cable Weight	For 06F -230 $\pm$ 10% Kg/Km For 48F -235 $\pm$ 10% Kg/Km For 96F -270 $\pm$ 10% Kg/Km
Impact Resistance (IEC 60794-1-21-E4)	25 Nm	Packing Length	4.0 Km $\pm$ 5%
Torsion Test (IEC 60794-1-21-E7)	$\pm 180^\circ$	Temperature Performance : Operation Installation Storage	-40°C to +70°C -30°C to +70°C -40°C to +75°C
Cable Bend Test (IEC 60794-1-21-E11)	Installation - 20 x Cable Diameter Operation - 10 x Cable Diameter		
Water Penetration Test (IEC 60794-1-22-F5B)	2 Meter/3 Meter/24 Hrs		

ORDERING INFORMATION

Description

Part No.

**Datacom SM 06F CLT Gel Filled Double Corrugated Steel Tape Armoured,
Double Jacket,PE ITUG657 A2 Cable**

DC-FO9A2CLTDTP-006

**Datacom SM 12F CLT Gel Filled Double Corrugated Steel Tape Armoured,
Double Jacket,PE ITUG657 A2 Cable**

DC-FO9A2CLTDTP-012

**Datacom SM 48F MLT Gel Filled Double Corrugated Steel Tape Armoured,
Double Jacket,PE ITUG657 A2 Cable**

DC-FO9A2MLTDTP-048

**Datacom SM 96F MLT Gel Filled Double Corrugated Steel Tape Armoured,
Double Jacket,PE ITUG657 A2 Cable**

DC-FO9A2MLTDTP-096